Beyond Language Barriers
Evaluating the Efficacy of Open-Source LLMs in Analyzing Sentiment in Non-English Textual Data

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Abstract

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1. Introduction

The rise of large language models (LLMs) has transformed natural language processing across domains and applications. While these models demonstrate impressive capabilities in English, their cross-linguistic effectiveness remains a critical area of investigation. This paper addresses a fundamental question in this domain: Can open source general purpose foundational LLMs accurately evaluate sentiment in non-English texts?

Recent research has uncovered a compelling phenomenon: prompting in English often improves performance in cross-linguistic tasks, even when the target language is non-English. Jiang et al. (2019) documented increased accuracy in knowledge extraction when using English prompts compared to diversified manual prompts in other languages. This finding was corroborated by Lin et

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al. (2021), who observed that English prompts outperform multilingual configurations across natural language understanding and machine translation tasks in 30 languages, even surpassing established baselines like GPT-3.

This pattern extends beyond isolated studies. Muennighoff et al. (2023), Nie et al. (2024), Tu et al. (2022), and Zhang et al. (2023) have all reported significant performance gains—measured by accuracy, F1 scores, and BLEU metrics—when models are guided with English instructions, regardless of the target language. These consistent findings suggest a fundamental characteristic of current LLM architectures that merits deeper investigation.

The primary purpose of this research is to determine whether untrained foundational models are adequate for analyzing textual sentiment in non-English contexts. While specialized models like BERT and its variants have been fine-tuned specifically for sentiment analysis tasks and perform effectively, they often require technical expertise to implement and adapt. By contrast, using non-fine-tuned foundational models could provide a more accessible solution for non-technical social science researchers who may lack the computational resources or programming knowledge necessary for model customization. This accessibility consideration is particularly relevant as AI tools become increasingly integrated into research methodologies across disciplines.

Sentiment analysis offers an ideal test case for examining this phenomenon, as it requires nuanced language understanding beyond simple translation. While proprietary models have received substantial attention in multilingual contexts, open source LLMs present both unique challenges and opportunities. They offer greater transparency, adaptability, and accessibility for global researchers and practitioners, yet their cross-linguistic capabilities remain less comprehensively documented.

This study systematically evaluates the performance of leading open source foundational LLMs in sentiment analysis across diverse non-English languages. We examine how prompt language affects performance, identify cross-linguistic patterns, and analyze model behavior across language families with varying resource availability. Our findings contribute to a deeper understanding of LLM capabilities and limitations in multilingual contexts and offer practical guidelines for optimizing their deployment in diverse language settings.

2. Research Question

Can open source general purpose foundational LLMs accurately evaluate sentiment in non-English texts?

3. Data & Methods

4. Results

5. Discussion

References